

AI-Based Image Colorization:

Deep Learning Improvement on Black & White Photographs

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Agenda

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Introduction

Objective

Develop a GAN-based solution to colorize black-and-white photographs automatically.

Highlights

- Preserves fine detail
- Produces vivid and accurate color predictions
- Scalable for historical restoration and digital media

Problem Statement

Problem Statement

Restore and enhance grayscale images by predicting and applying plausible colors.

Applications

- Digital art and archival restoration
- Film and media industry
- Cultural storytelling



Dataset Description

1 Dataset

Source: Kaggle (DIV2K)

2 Dataset Size

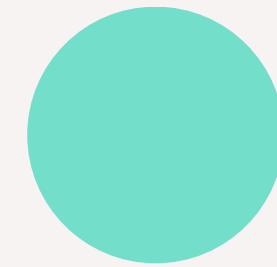
Size & Format: 800+ high-resolution PNG images

3 What's in dataset

Content: Landscapes, people, historic landmarks, objects

4 Preprocessing

- Cleaning corrupted files
- Normalization to [0,1]
- Data augmentation (flips, rotations)



Problem-Data Alignment

DIV2K's diverse and high-res content is ideal for learning intricate colorization patterns.

Suitability:

- Diverse image categories ensure better generalization
- High-resolution content preserves fine detail

Model Architecture

Model

- Model: Pix2Pix Conditional GAN (cGAN)
- Generator: U-Net architecture
- Discriminator: PatchGAN

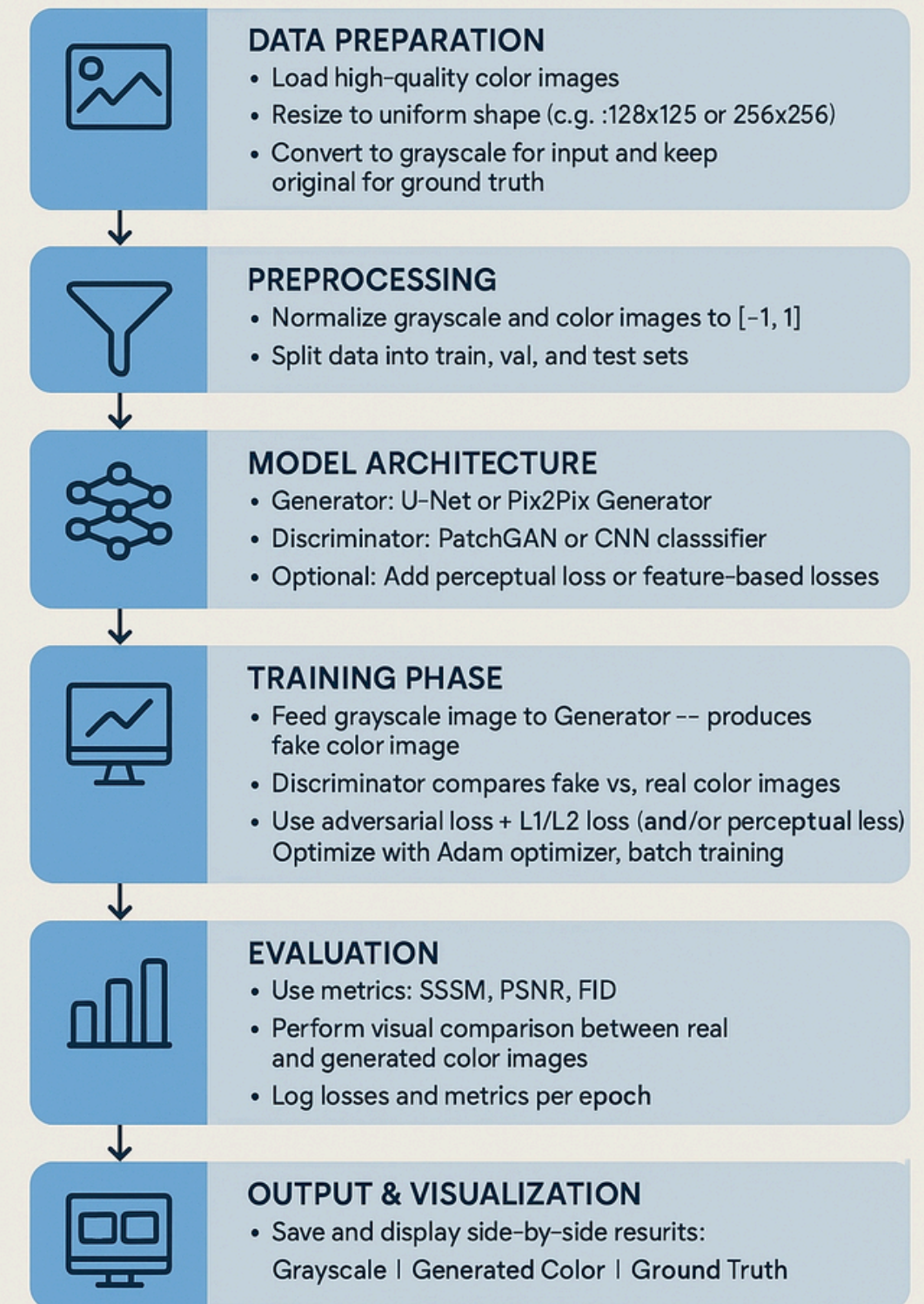
Loss Functions

- L1 Loss (pixel similarity)
- Binary Crossentropy (realism)
- FID considered for visual fidelity

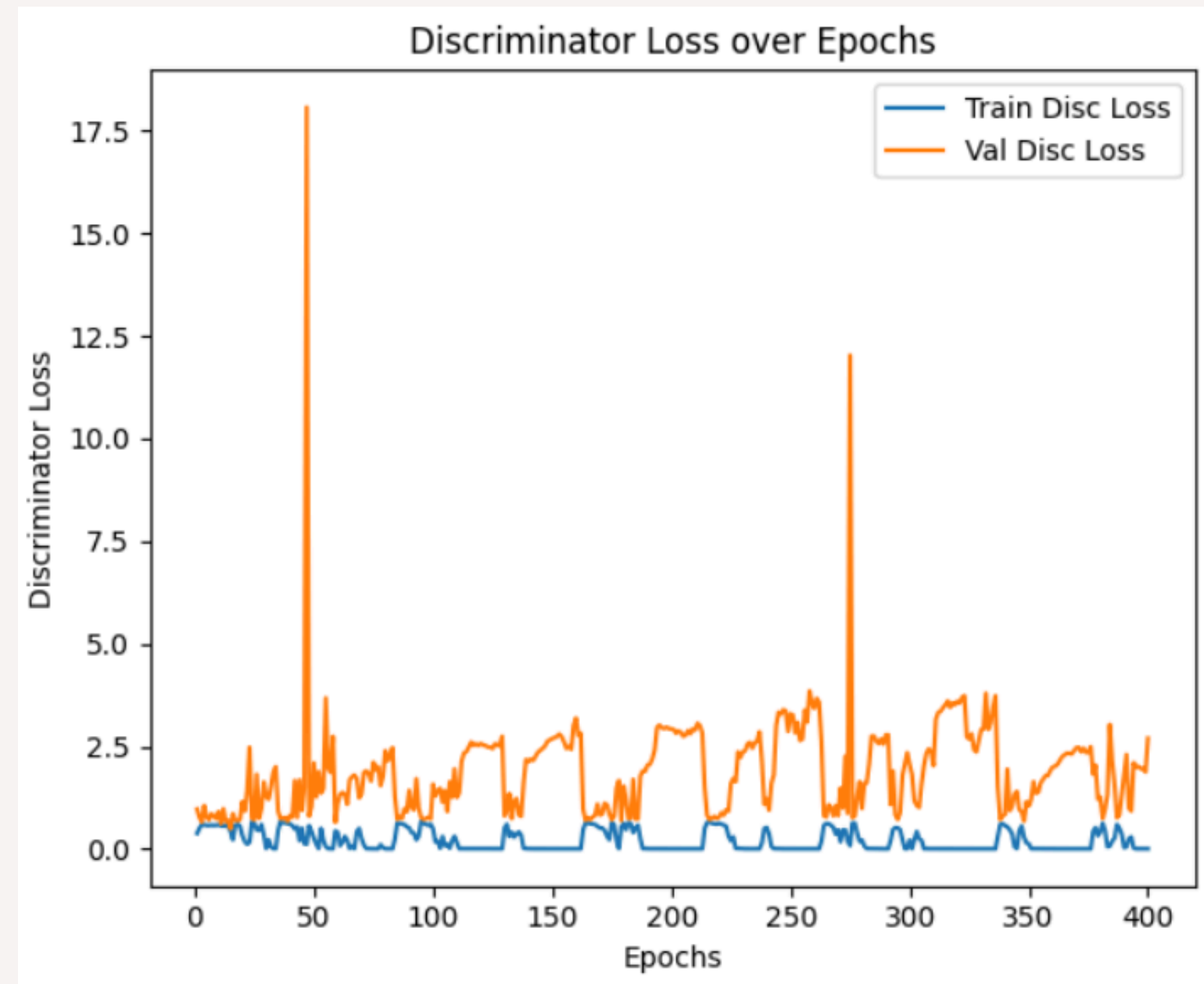
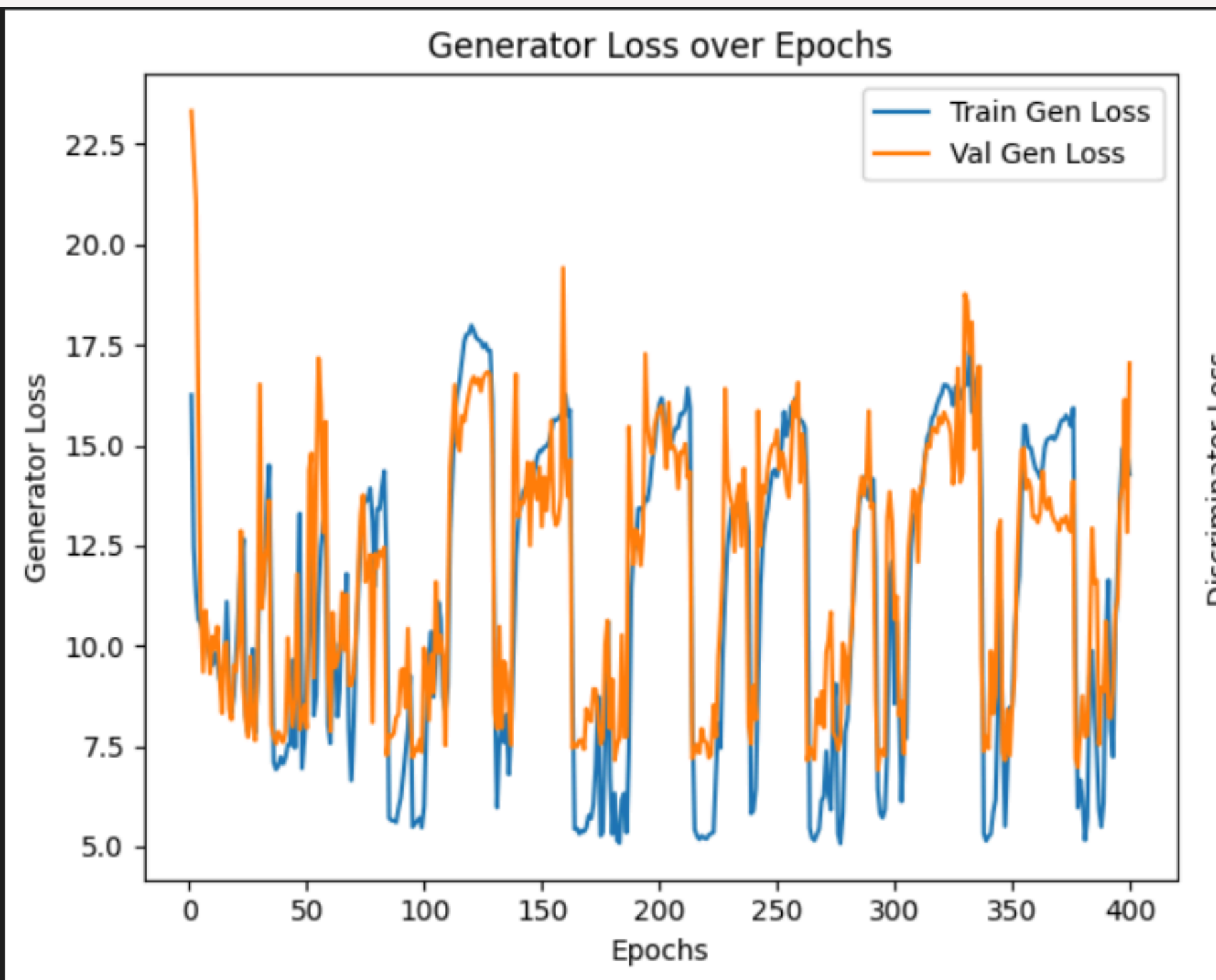


Implementation Pipeline

- Data Preparation
- Preprocessing
- Model Architecture
- Training Phase
- Evaluation
- Output & Visualization



Discriminator and Generator Loss



Evaluation Metrics

1

PSNR (Peak Signal-to-Noise Ratio): Measures pixel accuracy

2

SSIM (Structural Similarity Index): Measures visual similarity

3

FID (Frechet Inception Distance): Measures realism in feature space

Sample Results

Flow

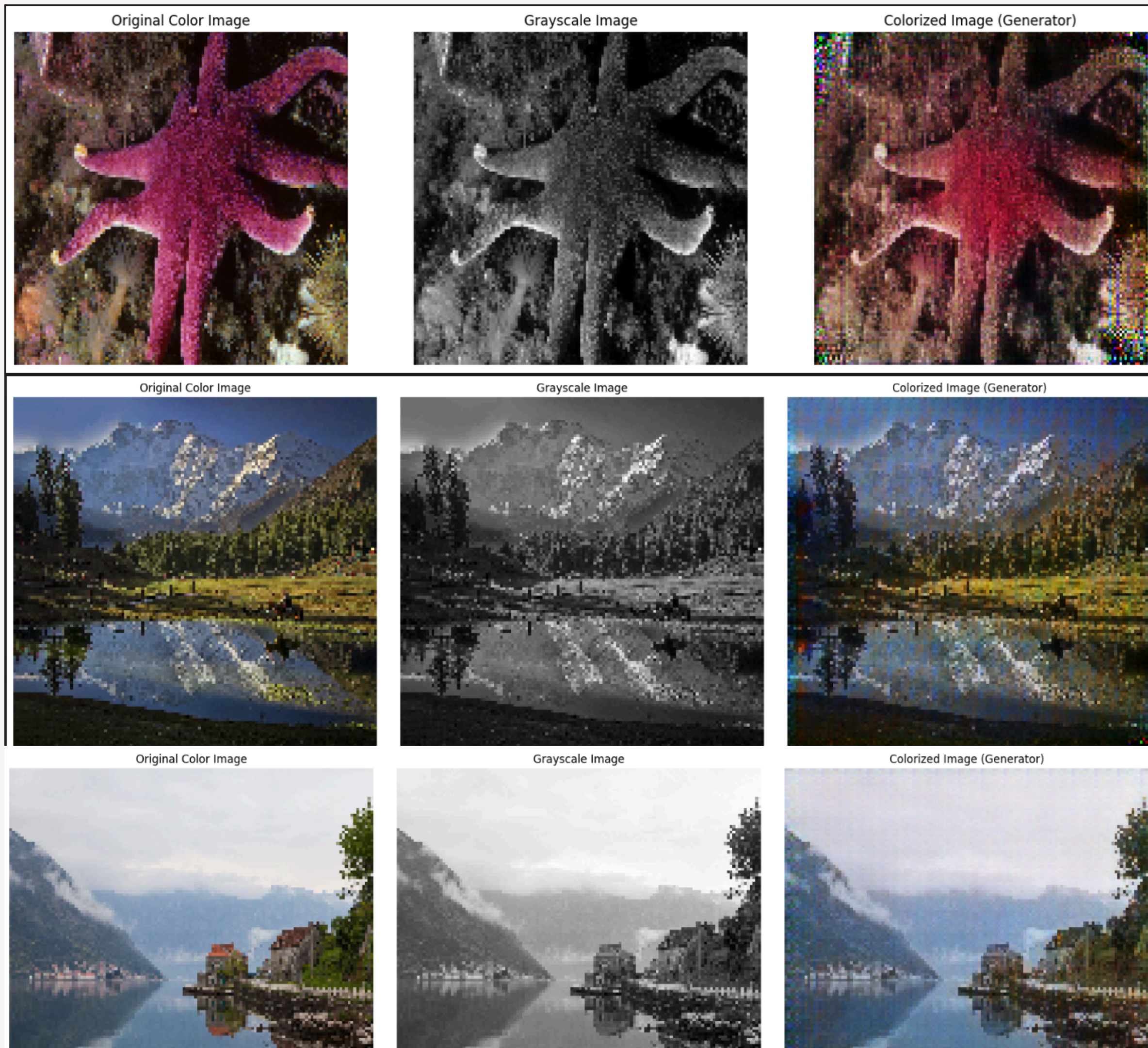
- Grayscale ➡ Generated ➡ Ground Truth

Epoch

- Epoch 100 Example:
- SSIM: 0.61
- PSNR: 21.3 dB

Quality

- Visual quality improves with training



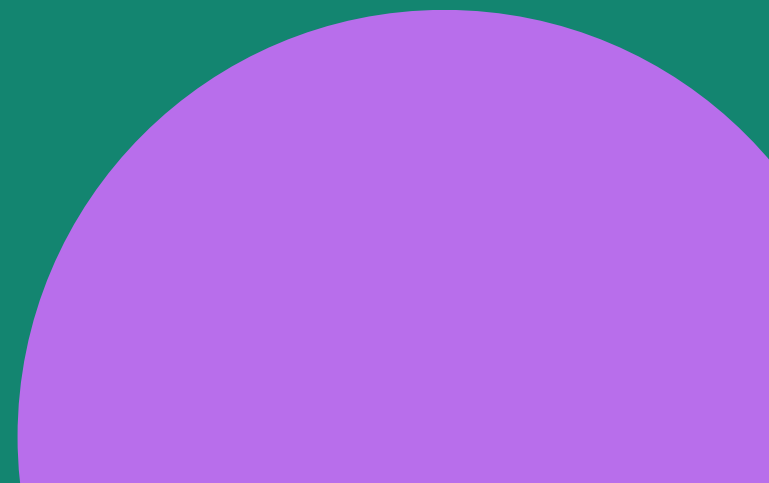
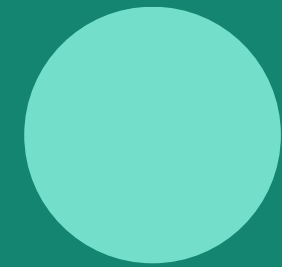
Challenges & Fixes

Preview executed before training in early iterations

Fixed: Corrected execution flow and denormalization logic

Initial outputs showed rainbow noise due to normalization errors

Fixed: Cleaned grayscale pipeline, restructured training order



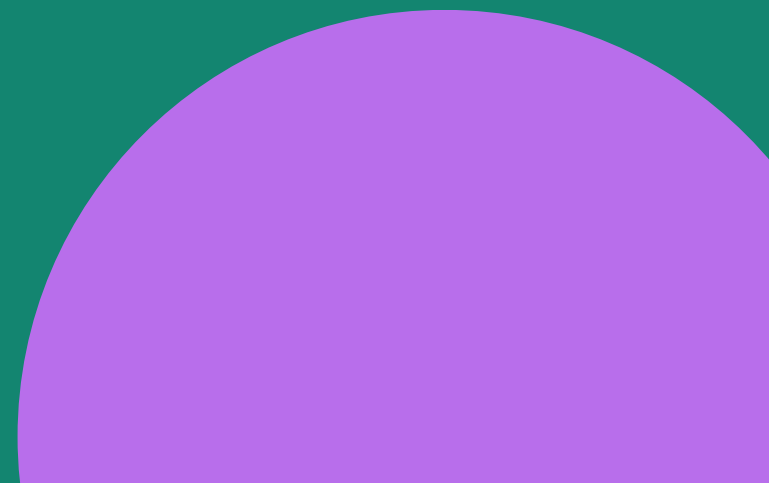
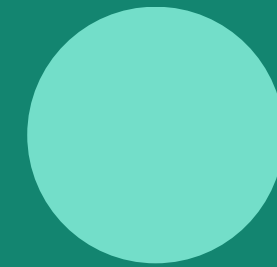
Ethical Considerations

- Risk of misrepresenting historical accuracy
- Dataset bias can influence color predictions
- Can be misused in deepfake scenarios if unchecked
- Addressed through careful dataset curation and academic use only



Conclusion & Q&A

- Developed an AI model that effectively colorizes grayscale photographs
- Balanced realism, structure, and diversity using Pix2Pix cGAN
- Future scope: Deploy as a web tool, train on historic datasets
- Thank you for your attention — we welcome your questions!



Thank

You

